

Dikshant

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EDUCATION

University of Alberta (UofA)

Edmonton, Canada

Masters in Computing Science

(Aug '24 - Present)

Supervisors: [Matthew Taylor](#) | [Marlos C. Machado](#)

Thesis: Planning with Laplacian Representation

Research Interests: Reinforcement Learning, Autonomous Driving and Robotics

Indian Institute of Technology Bombay (IITB)

Mumbai, India

Bachelors of Technology in Civil Engineering

(Jul '18 - May '22)

With Minor in Electrical Engineering

Department Rank 5 in a batch of 112 students

WORK EXPERIENCE

Data Driven Process Control

(Nov '24 - Feb '25)

Advisor: Payam Mousavi, Alberta Machine Intelligence Institute (AMII)

AMII | Repsol Spain

- Designed and implemented data-driven forward models for complex non-linear process control systems
- Developed predictive modeling approaches for industrial chemical processes without first-principles equations
- Benchmarked advanced control strategies including RL algorithms (PPO, SAC), DMDc, NMPC and PID
- Optimized distillation column and CSTR performance through comparative analysis of multiple control approaches

Infrastructure Cooperative Safety Assist System

(Oct '22 - Aug '24)

Advisor: Mr. Tokitomo Ariyoshi, Chief Engineer, CS Domain

Honda R&D Ltd. Japan

- Spearheaded the project to enhance safety of Honda's self-driving car in **occluded regions** using infrastructure cameras
- Modeled traffic-user trajectories with the intelligent driver model (**IDM**) implemented using Frenet Coordinates system
- Optimized (**2x faster**) and deployed the Collision Risk Map algorithm to proactively alert VRUs to potential collisions
- Implemented **3D-Net** model for monocular object detection to obtain the position and velocity of traffic participants
- Established a remote control room to access live CCTV streams utilizing python socket library over **4G** networks
- Simulated collision detection on Gazebo and employed advanced **HMI** techniques to alert users of potential collisions

Meta Learning

(Jun '22 - Aug '22)

Advisor: Prof. Matthew Taylor, Intelligent Robot Learning Lab

University of Alberta, Canada

- Implemented Model Agnostic Meta Learning (**MAML**) on a multi-task SinWave regression using the **JAX** framework
- Replaced iterative learning algorithm with Markov Reward Process for learning approximate meta value functions
- Conducted experiments to comprehend the reason behind **poor memory scaling** on increasing gradient descent steps

Camera and Radar Fusion

(May '22 - Jul '22)

Advisor: Mrs. Misa Komuro, Assistant Chief Engineer, CS Domain

Honda R&D Ltd. Japan

- Merged camera and radar-based object detection results to improve the traffic user detection efficiency by **20%**
- Applied temporal and spatial alignment techniques to **synchronize** the sensor inputs, increasing the tracking precision
- Visualized the fusion results on **RViz** using ROS Visual Markers and tested robustness of algorithm in crowded regions

Scalable Multi-Agent RL Training School for Autonomous Driving - SMARTS

(May '21 - Aug '21)

Advisor: Prof. Matthew Taylor, Intelligent Robot Learning Lab

University of Alberta, Canada

- Constructed a training pipeline in SMARTS for multiple ego agents across varying map and traffic-pattern complexities
- Trained single ego-agent scenarios using **PPO** over increasingly difficult maps and traffic, **modelling human learning**
- Initialised multiple ego-agent training with single-agent weights, obtaining **3x faster convergence** over a fixed horizon
- Enhanced T-intersection safety by training agents with additional **hand-coded** positional data shared across neighbours

Multi-Agent Patrolling

(Apr '20 - Apr '21)

Advisors: Prof. Arpita Sinha and Prof. Leena Vacchani, Department of Systems & Control

IIT Bombay

- Devised an **IoT**-based multi-robot patrolling algorithm using **Deep Q-Network** for effective monitoring in remote areas
- Created an interface for communication between the agents in **SUMO** by using Traffic Control Interface (**TraCI**) library

- Formulated a **decentralized junction-based** decision making approach, reducing the junction visit intervals by **2x**
- Engineered sensor-failure resilience by thresholding node idleness; beat SOTA in topologies with large failures by **8%**
- Co-authored a research **manuscript** titled as "*Interconnecting Vehicles using IoT Framework for Multi-Agent Patrolling*"

Room Service Automation Bot

(Jun '20 - Jul '20)

Golden Oak Projects Private Limited

Tech startup

- Developed an autonomous hotel service robot adept at delivering amenities and responding to room service orders
- Implemented **SLAM** for real-time mapping of the environment by integrating the data from encoder, IMU and 2D-Lidar
- Simulated the scenarios on a custom-made Gazebo environment and utilized **DWA** algorithm for efficient navigation

TEACHING ASSISTANT

CMPUT 503: Experimental Mobile Robotics

(Winter '25)

Department of Computing Science, University of Alberta

An introductory course to experimental inquisition of the fascinating field of mobile robotics. It covers basic concepts, models, and algorithms in autonomous navigation of a mobile robot operating in indoor environments. Using ROS, it involves robot mapping, localization, path planning, object detection and homing.

CMPUT 366: Search Planning in Artificial Intelligence

(Fall '24)

Department of Computing Science, University of Alberta

An introductory course to search and planning in artificial intelligence. It covers deterministic single-agent and multi-agent problems; how to model real-world problems as state-space search problems and how to solve such problems. It covers algorithms for solving deterministic shortest path problems with combinatorial optimization problems, constraint satisfaction problems, and multi-agent problems.

MA108: Differential Equations

(Spring '22)

Department of Mathematics, Indian Institute of Technology Bombay

An introductory differential equations course covering first-order and second-order differential equations, along with their applications. The course focuses on solving linear and nonlinear ODEs using various methods like separation of variables, integrating factors, and series solutions.

TECHNICAL PROJECTS

SeDriCa | Unmesh Mashruwala Innovation Cell

(Aug '20 - Apr '22)

Worked in a **20+** member team aiming to develop a **Self-Driving Car** capable of traversing unstructured environments

- Designed a hybrid **MPC** controller using **dynamic bicycle model** to attain higher speeds and tackle adverse scenarios
- Crafted a drive-by-wire (**DBW**) system from ground up for mapping accelerator pedal position to DBW throttle body
- Integrated sub-modules for real-time deployment using **CAN** module to establish communication with the hardware
- Performed end-to-end simulation for the entire setup from perception to control on a self-made Gazebo environment
- Implemented **RRT*** and **Hybrid A*** for path planning and optimized the route by integrating data from Google Maps

Intelligent Picking Robot | Flipkart Grid 2.0-Robotics Challenge

(Jun '20 - Aug '20)

Led a team of 5 developing a robotic arm capable of picking and transporting items in a warehouse | **Top 2%** across India

- Designed a **4-DoF** robotic manipulator and visualized its grasping mechanism on **RViz** using the **MoveIt** framework
- Implemented **YOLOv3** algorithm for object detection and identified potential grasping points using **OpenCV** functions
- Mapped the area between pick and drop stations using **SLAM** and employed **A*** algorithm for efficient path planning

Terrace Farming Bot | Inter-IIT Technical Meet 8.0

(Oct '19 - Dec '19)

Worked in a team of 8 to develop an autonomous step-climbing robot to perform terrace farming operations

- Employed a linear actuator mechanism using stepper motors and developed **CAD** models for precise manufacturing
- Fused monocular **visual odometry** and data from MPUs and stepper motor encoders for efficient state estimation
- Deployed a **PID**-based position and orientation controller employing ultrasonic sensors to provide accurate feedback

Parallelizing A* and D* Algorithms | High Performance Scientific Computing

(Feb '21 - May '21)

Advisor: Prof. S. Gopalakrishnan, Mechanical Engineering Department

Course Project

- Enhanced the computational efficiency of **A*** and **D*** by parallelizing them with **OpenMP** and **CUDA** frameworks
- Demonstrated and visualised improved path planning of parallel **A*** by implementing it using **pygame** library

Autonomous Quadruped Robot | RoboCup Rescue League Challenge

(Sep '19 - May '20)

Part of **17** member team working on autonomous quadruped for its easy maneuvering on any terrain

- Performed **Inverse Kinematic** analysis for quadruped control to ensure an optimum balance on uneven ground
- Studied rhythmic movement of each leg of the quadruped for optimal **gait selection** based on speed and terrain

OTHER TECHNICAL ACTIVITIES

- Delivered **talks** on RL for autonomous driving and Control Theory to **70+ students** with demonstrations ('20)
- Guided **150+** freshmen teams in XLR8 competition to build and debug a wireless bluetooth controlled bot ('19)
- Replicated **Thor's Hammer** using electromagnetism and RFID for Institute Technical Council Orientation ('19)
- Designed a **line-follower robot** capable of solving a complex maze in the Institute's technical fest competition ('19)
- Assembled a **bluetooth-controlled robot** and applied differential-drive mechanism for optimum control ('19)
- Developed **Xylobands** to represent the robotics club in the Institute Technical Orientation for the freshmen batch ('19)

SCHOLASTIC AND TECHNICAL ACHIEVEMENTS

- Acquired **N5** level of certification in the Japanese Language Proficiency Test (**JLPT**) ('20)
- Conferred with the **Department Technical Excellence Award**, given annually to **1** in **400** students ('22)
- Recipient of the Kishore Vaigyanik Protsahan Yojana (**KVPY**) Fellowship awarded by the Government of India ('18)
- Secured **97.8 percentile** in JEE Advanced entrance examination among **0.15** million+ candidates ('18)

TECHNICAL PROFICIENCY

Languages	Python, C++, Matlab, OpenCL, Octave, Bash, CSS, Markdown, TeX
Softwares	Webots, Gazebo, CARLA, SUMO, OSM, Netedit, TraCI, Simulink, Auto-CAD
Frameworks/Libraries	JAX, PyTorch, Tensorflow, RLlib, ROS, Gym, Sockets, Optax, CV, Pygame
Electronics	Jetson Orin, Raspberry Pi, Arduino, Node MCU, ESP32

LEADERSHIP AND MENTORSHIP ROLES

Team Leader | SeDriCa, UMIC

(May '21 - May '22)

Led a 26-member team representing IIT Bombay at the International Ground Vehicle Competition

- Managed operations, finances, logistics and knowledge-transfer in a **4-tier, multi-disciplinary** student technical team
- Setting vision and strategy for the mechatronics subsystem | Spearheaded exploration of **RL-based control** methods
- Raised **Rs 3.5 M** from IITB and Mahindra RISE, and forged relations with multiple academic and industry experts

Technical Head | Inter-IIT Tech Meet 9.0

(Jan '21 - Mar '21)

Part of a 50-member strong contingent working on 10 projects of medium and high-level difficulties

- Conducted **40+** stringent interviews for selecting contingent managers, project leads, sponsors and financial advisors
- Provided critical technical support for resolving the issues in docker, ROS, gazebo, and integration in various projects

Convener | Electronic and Robotics Club, Institute Technical Council

(May '19 - Apr '20)

Part of a 15+ member team that conceptualizes and organizes events for tech enthusiasts in the Institute

- Hosted a widely-attended **ROS workshop** aimed at beginners during the COVID, drawing over **300 tech enthusiasts**
- Spearheaded XLR8, handling **600+** participants (up **30%**), achieving highest bot-completion rate (**92%**) in four years
- Conducted tutorial sessions on Arduino, Raspberry Pi, Image Processing, and **Serial Communication Protocols**
- Ideated and conducted '**Jhatka GC**,' an electronics and robotics puzzles-based inter-hostel championship

KEY COURSES UNDERTAKEN

Computer Science	Deep Reinforcement Learning, Optimization and Decision Making under Uncertainty, Foundation of Intelligent & Learning agents, Artificial Intelligence and Machine Learning, Deep Learning Specialization, Introduction to Machine Learning, Automatic Speech Recognition, Computer Programming and Utilisation
Controls and Robotics	Autonomous Driving Motion Planning & Coordination of Autonomous Vehicles, Linear and Non-linear systems, Introduction to Robotics, Intelligent Feedback & control systems, Adaptive Control theory, Linear Algebra, Differential Equations, ROS: Localization, Navigation and SLAM, High Performance Scientific Computing
Electrical	Probability and Random Processes, Power Electronics, Signal Processing
Interdisciplinary	Calculus, Differential Equations, Quantum Physics, Psychology, Economics

EXTRACURRICULARS

Sports	• Represented Computer Science Graduate Student Association (CSGSA) of UofA in winter Volleyball Intramurals, 2024 and summer Soccer Intramurals, 2025 • Participated in Annual Tokyo Midtown Futsal Tournament 2023, representing Honda • Secured 1st place in the Institute's annual inter-hostel football & cricket tournaments 2021 • Represented IITB in Football, Cricket and Athletics at the Annual Training Camp 2019
Volunteering	• Served as an NCC cadet in the 2-Maharashtra Engineering Regiment, 2018-19 contingent • Selected as cadet for the Republic Day Parade '19; awarded with NCC ATC certificate • Counseled students from local municipal schools for IITB's Career Counseling Campaign
Miscellaneous	• Attended Enhanced Safety of Vehicles Conference (ESV) 2023 through Honda R&D • Bagged 1st position in the Logic GC organized by the Maths and Physics Club of IITB • Awarded Hostel Technical Special Mention for exemplary contribution to institute tech